Zinal Jain

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academic-portfolio-ten.vercel.app

Education

G H Patel College of Engineering & Technology 2023 - 2027 (expected)

B.Tech in Computer Science Engineering (IoT) · Minor in Robotics
CGPA: 8.56/10 · Anand, Gujarat

Research interests

UAV navigation, embedded perception, and edge AI for robotic and IoT systems.

Research experience

Research Intern · DRDO, Defence Technology & Test Centre Aug 2026 - Present

Lucknow · Edge AI-Based Human Detection and Alert System for Perimeter Surveillance

- Built and tested a laptop pipeline: motion detection, MobileNet-SSD V2 on ONNX Runtime, multi-frame confirmation, and event recording.
- Proposed a Raspberry Pi, camera, EBYTE E22 LoRa, and ESP32 hardware setup for future implementation.

Selected projects

Neerkavach · Semi-autonomous water rescue drone 2026 - ongoing

Team project · National Finalist, GUJCOST Robofest 6.0 Hackathon · Team funding: ₹50,000

- Team built a working water rescue drone prototype designed to fly to a person in distress, land on water, and move a flotation tube toward them using two thrusters.
- **My work:** Flight-controller calibration and setup, waypoint mission planning, and semi-autonomous flight operation from a laptop ground station.
- **Status:** Waypoint missions and water landings tested; drowning-person detection validated in simulation. Depth-camera landing and onboard detection remain in development by the team.

ARCNET · Aerial Robotics Minefield Navigation 2025

Team project · National Finalist, GUJCOST Robofest 5.0 · Team funding: ₹2,50,000

- Worked on the Raspberry Pi 4-Pixhawk MAVLink connection, troubleshot connection settings, and set up ArduPilot for waypoint navigation; verified live telemetry and command acknowledgements.
- Flight operations later used a telemetry-radio connection with a laptop ground station. The team produced a working drone navigation prototype.

KRAKEN · Underwater remotely operated vehicle 2024

Team project · National Finalist, GUJCOST Robofest 4.0 · Team funding: ₹2,50,000

- Worked on documentation and helped with wiring and assembly. The team built a working eight-thruster, six-degree-of-freedom underwater robot.

SSIP Competition: Second Prize and ₹10,000 in project funding.

Technical skills

Programming: Python, C/C++ · **Platforms:** Raspberry Pi, Pixhawk
Robotics and tools: ArduPilot, MAVLink, Mission Planner, QGroundControl

Certifications

SAP Code Unnati - AI/ML & Edge Computing · Ethical Hacking - SVNIT Surat